Harsh Gupta

Roll Number: 22IM30014
B.Tech - M.Tech Integrated
Industrial and Systems Engineering
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Education

- **Indian Institute of Technology Kharagpur** *2022 - 2027*
  Dual Degree, Industrial and Systems Engineering

Professional Experience

- **Practworks** *Dec 2023 - Mar 2024*
  Machine Learning Engineer Intern
  – Constructed a comprehensive dataset and **concept dependency graph**, establishing inter-concept relationships
  – Developed a robust **synthetic data generation algorithm** (48,000 data points) to address data scarcity
  – Led rigorous testing and fine-tuning, achieving **67% accuracy** and improving model performance
- **Ylytic** *Sep 2023 - Nov 2023*
  Software Engineer Intern
  – Integrated **5 new data features** into the company's database, enhancing data-driven insights and decision-making
  – **Automated task execution** and implemented rigorous testing protocols, ensuring data processing and reliability
  – Optimized workflows, **reduced processing time(by 60%)**, and ensured error-free operations for **2.4 million data**

Projects

- **Azad Hall Website** *Oct 2023 - Feb 2024*
  Modernized Azad Hall's website with features and performance for seamless user interactions
  – Deployed **Django website on GCP** with Nginx, Gunicorn, and Docker for scalability and maintenance
  – Architected website to handle **700+ concurrent users**, ensuring website responsiveness and performance
  – Optimized **PostgreSQL database in GCP** for efficient data management, enabling seamless website features
- **Self-Evaluation of Academic Performance** *Jun 2023 - Oct 2023*
  Harnessed the power of smartphone usage data to build a novel model for predicting student GPA Research Project
  – **Preprocessed smartphone data**, cleaning and analyzing features to identify high-quality features
  – Evaluated regression models (**Lasso, Ridge, SVR**) using **K-fold cross-validation** for unbiased performance
  – Optimized predictive performance with Ridge regression, achieving a **Mean Absolute Error of 0.255**

Competitions

- **Bhashini Sprint | Voice and Image Controlled Inventory Management** *Mar 2024 - Present*
  Built voice-controlled inventory system, selected as a top 3 team in the Bhashini Sprint competition
  – Developed an **audio recording system** with Bhashini ASR integration, enabling **voice-based inventory input**
  – **Fine-tuned** the Gemini 1.0 Pro 001 for precise **text-to-command conversion**
  – Designed and implemented MySQL database to support **efficient inventory management** via a custom algorithm

Skills

Languages: Python, C++, C **Developer Tools**: Git (Version Control), PyCharm, VS code, Docker, Postman, Figma **Frameworks and Libraries**: Flask, Scikit-learn, Hugging Face, NLTK, spaCy, Matplotlib, Pandas, Numpy **Cloud/Databases**: MongoDB, AWS, GCP

Relevant Coursework

Programming and Data Structures, Algorithm Lab, Operation Research lab, Probability and Statistics, Linear Algebra, Numerical and Complex Analysis, Operations Research, Work System Design, Economics

Positions of Responsibility

– **Associate Quant Researcher** Quant Club, IIT Kharagpur *May 2023 - Present*
  ∗ Developed **algorithmic trading strategies**, applying knowledge gained in the Summer of Quant program
  ∗ Fostered a collaborative learning environment within Quant Club, **promoting skill development and engagement**

Extra-Curricular Activities

– **Volunteer,** National Service Scheme *Nov 2022 - Apr 2024*
  ∗Spearhead community development projects, fostering sustainable solutions for social issues
  ∗Design and execute engaging workshops and outreach programs to promote community involvement